

Total No. of Questions : 8]

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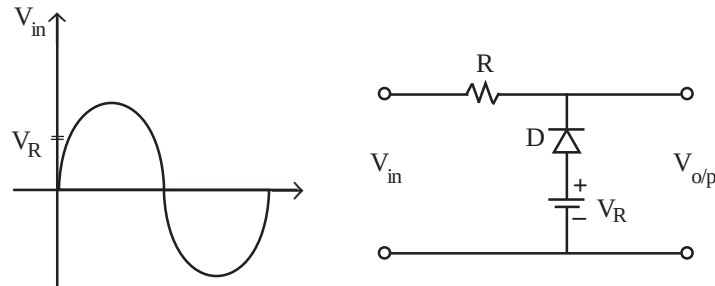
Roll No

EC/EX-304 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Electronics Devices
Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.

1. What is Clipper Circuit? Draw output of following circuit.



2. Determine the diffusion constants of electrons and holes of a Silicon single crystal at 27°C, if the mobilities of electrons and holes are 0.17 and 0.025 m²/v-sec resp. at 27°C.
3. Write short notes on any two :
- i) LED
 - ii) h-parameters
 - iii) Early effect
 - iv) UJT
4. What will happen to the Zener breakdown and avalanche breakdown voltages when the temperature is increased? Give answer with the help of suitable diagram.
5. Define active, saturation and cutoff mode of operations of a transistor.
6. Draw equivalent circuit and characteristic of tunnel diode also explain tunneling effect.

OR

- a) Discuss Ebers-Moll model of transistor in common Emitter configuration for a PNP transistor.
- b) Given $I_E = 2.5\text{mA}$, $h_{fe} = 140$, $h_{oe} = 20\mu\text{s}$ and $h_{ob} = 0.5\mu\text{s}$. Find the CE hybrid equivalent circuit.

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7. Describe the mechanism by which light increase the conductivity of a bulk type of a photoconductive cell. Give three characteristics of photoconductive cell.

OR

- a) Draw and explain VI characteristics of PN junction diode.
 - b) Define transition and diffusion capacitances found in diode.
8. Write short notes on any two of the following:
- a) Avalanche diode
 - b) PIN diode
 - c) Photo transistor
 - d) Miller effect
